Stanisław Szufa

Publication List

Summary. 5 journal articles and 28 conference papers, published mainly at the major AI conferences (AAAI, IJCAI, NeurIPS, AAMAS) and in *Artificial Intelligence*, *JAIR*, and *JCSS*.

Author ordering. As is customary in computational social choice, authors are usually listed alphabetically, so position in the author list does not reflect contribution. Papers that deviate from alphabetical order list the lead author first.

Journal Articles

[J5] P. Skowron, A. Slinko, **S. Szufa**, and N. Talmon. Participatory Budgeting with Cumulative Votes. *Theory and Decision*, 2025.

[J4] **S. Szufa**, N. Boehmer, R. Bredereck, P. Faliszewski, R. Niedermeier, P. Skowron, A. Slinko, and N. Talmon. Drawing a Map of Elections. *Artificial Intelligence* 343, 2025. Journal version of AAMAS-2020 and IJCAI-2021 papers.

[J3] P. Faliszewski, P. Skowron, A. Slinko, K. Sornat, **S. Szufa**, and N. Talmon. How Similar Are Two Elections? *Journal of Computer and System Sciences* 150, 2025. Journal version of the AAAI-2019 paper.

[J2] P. Faliszewski, K. Sornat, and **S. Szufa**. The Complexity of Subelection Isomorphism Problems. *Journal of Artificial Intelligence Research* 80, pp. 1343–1371, 2024. Journal version of the AAAI-2022 paper.

[J1] N. Boehmer, K. Heeger, and **S. Szufa**. A Map of Diverse Synthetic Stable Matching Instances. *Journal of Artificial Intelligence Research* 79, pp. 1113–1166, 2024. Journal version of the AAMAS-2023 paper.

Conference Papers

[C28] R. Faulkner, **S. Szufa**, D. Hoyer, T. J. Zhang, R. Bouffanais, J. Z. Leibo, and Z. Jin. Simulating Democratic Deliberation: Representing Pluralistic Preferences through Electoral Systems in Multi-Agent LLMs. *Findings of EMNLP-2026*, pp. 11673–11705.

[C27] P. Faliszewski, J. Mertlová, K. Sornat, **S. Szufa**, and T. Wąs. Agreement, Diversity, and Polarization Indices for Approval Elections. *Proceedings of IJCAI-2026*, pp. 3393–3401.

[C26] P. Faliszewski, K. Sornat, **S. Szufa**, and T. Wąs. Outer Diversity of Structured Domains. *Proceedings of AAMAS-2026*, pp. 2106–2114.

[C25] P. Böhm, R. Bredereck, P. Gölz, A. Kaczmarczyk, and **S. Szufa**. Putting Fair Division on the Map. *Proceedings of AAAI-2026*, pp. 16726–16734.

[C24] P. Faliszewski, K. Sornat, **S. Szufa**, and T. Wąs. Diversity of Structured Domains via k-Kemeny Scores. *Proceedings of AAAI-2026*, pp. 16880–16888.

[C23] P. Faliszewski, Ł. Janeczko, A. Kaczmarczyk, G. Lisowski, P. Skowron, **S. Szufa**, and M. Szwagierczak. Strategic Cost Selection in Participatory Budgeting. *Proceedings of NeurIPS-2025*.

[C22] F. Nikolow, P. Faliszewski, and **S. Szufa**. Maps of Tournaments: Distances, Experiments, and Data. *Proceedings of ECAI-2025*, pp. 675–682.

[C21] D. Boratyn, W. Słomczyński, D. Stolicki, and **S. Szufa**. Spoiler Susceptibility in Party Elections. *Proceedings of ECAI-2025*, pp. 3775–3782.

[C20] P. Faliszewski, Ł. Janeczko, A. Kaczmarczyk, M. Kurdziel, G. Pierczyński, and **S. Szufa**. Learning Real-Life Approval Elections. *Proceedings of AAMAS-2025*, pp. 704–712.

[C19] P. Faliszewski, J. Mertlová, P. Nunn, **S. Szufa**, and T. Wąs. Distances Between Top-Truncated Elections of Different Sizes. *Proceedings of AAAI-2025*, pp. 13823–13830.

[C18] T. Delemazure, Ł. Janeczko, A. Kaczmarczyk, and **S. Szufa**. Selecting the Most Conflicting Pair of Candidates. *Proceedings of IJCAI-2024*, pp. 2766–2773.

[C17] N. Boehmer, P. Faliszewski, Ł. Janeczko, D. Peters, G. Pierczyński, Š. Schierreich, P. Skowron, and **S. Szufa**. Evaluation of Project Performance in Participatory Budgeting. *Proceedings of IJCAI-2024*, pp. 2678–2686.

[C16] W. Słomczyński, D. Stolicki, and **S. Szufa**. Nonparametric Detection of Gerrymandering in Multiparty Plurality Elections. *Proceedings of IJCAI-2024*, pp. 2967–2975.

[C15] N. Boehmer, P. Faliszewski, Ł. Janeczko, A. Kaczmarczyk, G. Lisowski, G. Pierczyński, S. Rey, D. Stolicki, **S. Szufa**, and T. Wąs. Guide to Numerical Experiments on Elections in Computational Social Choice. *Proceedings of IJCAI-2024*, pp. 7962–7970. Survey track.

[C14] G. Pierczyński and **S. Szufa**. Single-Winner Voting with Alliances: Avoiding the Spoiler Effect. *Proceedings of AAMAS-2024*, pp. 1567–1575.

[C13] Ł. Janeczko, J. Lang, G. Lisowski, and **S. Szufa**. Discovering Consistent Subelections. *Proceedings of AAMAS-2024*, pp. 935–943.

[C12] P. Faliszewski, M. Lackner, K. Sornat, and **S. Szufa**. An Experimental Comparison of Multiwinner Voting Rules on Approval Elections. *Proceedings of IJCAI-2023*, pp. 2675–2683.

[C11] P. Faliszewski, J. Flis, D. Peters, G. Pierczyński, P. Skowron, D. Stolicki, **S. Szufa**, and N. Talmon. Participatory Budgeting: Data, Tools and Analysis. *Proceedings of IJCAI-2023*, pp. 2667–2674.

[C10] P. Faliszewski, A. Kaczmarczyk, K. Sornat, **S. Szufa**, and T. Wąs. Diversity, Agreement, and Polarization in Elections. *Proceedings of IJCAI-2023*, pp. 2684–2692.

[C9] N. Boehmer, K. Heeger, and **S. Szufa**. A Map of Diverse Synthetic Stable Roommates Instances. *Proceedings of AAMAS-2023*, pp. 1003–1011. Pragnesh Jay Modi Best Student Paper Award.

[C8] N. Boehmer, R. Bredereck, E. Elkind, P. Faliszewski, and **S. Szufa**. Expected Frequency Matrices of Elections: Computation, Geometry, and Preference Learning. *Proceedings of NeurIPS-2022*.

[C7] **S. Szufa**, P. Faliszewski, Ł. Janeczko, M. Lackner, A. Slinko, K. Sornat, and N. Talmon. How to Sample Approval Elections? *Proceedings of IJCAI-2022*, pp. 496–502.

[C6] N. Boehmer, P. Faliszewski, R. Niedermeier, **S. Szufa**, and T. Wąs. Understanding Distance Measures Among Elections. *Proceedings of IJCAI-2022*, pp. 102–108.

[C5] P. Faliszewski, K. Sornat, and **S. Szufa**. The Complexity of Subelection Isomorphism Problems. *Proceedings of AAAI-2022*, pp. 4991–4998.

[C4] N. Boehmer, R. Bredereck, P. Faliszewski, R. Niedermeier, and **S. Szufa**. Putting a Compass on the Map of Elections. *Proceedings of IJCAI-2021*, pp. 59–65.

[C3] **S. Szufa**, P. Faliszewski, P. Skowron, A. Slinko, and N. Talmon. Drawing a Map of Elections in the Space of Statistical Cultures. *Proceedings of AAMAS-2020*, pp. 1341–1349.

[C2] P. Faliszewski, P. Skowron, A. Slinko, **S. Szufa**, and N. Talmon. How Similar Are Two Elections? *Proceedings of AAAI-2019*, pp. 1909–1916.

[C1] P. Faliszewski, **S. Szufa**, and N. Talmon. Optimization-Based Voting Rule Design: The Closer to Utopia the Better. *Proceedings of AAMAS-2018*, pp. 32–40.

Preprints and Manuscripts

[P2] **S. Szufa**, D. Scherer, D. Stolicki, R. Bouffanais, and D. Wernli. Abstention Patterns in the United Nations General Assembly. *Manuscript*, 2026.

[P1] U. Grandi, R. Bredereck, T. Delemazure, U. Endriss, J. Maly, N. Mattei, N. Maudet, O. Nardi, and **S. Szufa**. Social Choice with Text: Collective Decision Making in the LLM Era. *Position paper, preprint, HAL hal-05548117*, 2026. Arising from a working group at Dagstuhl Seminar 25401.

Thesis

[T1] **S. Szufa**. *Map of Elections*. PhD thesis, AGH University of Kraków, 2024. Defended with distinction; Polish Prime Minister's Award for an outstanding doctoral dissertation.

Software and Datasets

[S1] **Pabulib** (pabulib.org). Open library of participatory budgeting elections; 2,053 instances from 30 cities in 8 countries, cited in more than 100 publications. Founded in 2020.

[S2] **prefsampling**. Python library for sampling synthetic preference data under a wide range of statistical cultures. Co-developed with Simon Rey.

[S3] **mapof**. Python library for computing distances between elections and drawing maps of elections.